Design Viability Description	Connection to Design Requirement	Credible Evidence
The system used an Arduino kit, a smoke sensor, and a water pump.	Must suppress fire within one minute	Unable to do final testing due to issues with the smoke sensor
Total cost remains under $50	All the materials that were used had to stay under the $50 budget limit. Materials that had been “donated” needed to be counted in as well.	Supply lost with prices showed cost of around $40
Water pump system outside of the structure with plastic tubes going around the ceiling	Must suppress the fire with no human external interactions.	Unable to do final testing due to issues with the smoke sensor
Design is powered externally, no batteries or explosives used	External power source required	System would run on a USB power source, no batteries or flammables included.
Structure includes required interior elements	Furniture, doorways and windows requirements specified.	Model included all the required furniture and structure features.
Sub systems such as the sensor, pump, and tubing are proven components	Must function reliably once it's assembled	All components had been researched previously to ensure they were compatible with each other.

To ensure the viability of our fire detection and suppression system, we evaluated our design decisions against each design requirement using credible evidence. Our main system was built around an Arduino and a smoke sensor with a water pump, which research has proven to trigger water dispersion in around a minute.
We also carefully chose materials that stayed under the budget limit, as well as choosing good quality materials.